

Part 4

(10 min)

PEP outputs: numerical vs. closed form/analytical

How to obtain mathematical proofs from a PEP?

Outline

PEP outputs: numerical vs. analytical/closed form

Proofs from dual multipliers

Complete worst-case of gradient method for smooth convex functions

What you obtain when you solve a PEP

1. In all cases:
 - numerical value for worst-case performance
 - numerical description of worst-case function (interpolate primal)
 - numerical proof of worst-case performance (dual multipliers)

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2. With extra work: explicit formulas for some/all of above
→ this requires fitting/guessing algebraic expressions in N
3. In the best case: a rigorous mathematical proof
→ this requires a proof with (possibly tedious) reformulation of worst-case bound as an equality involving sums-of-squares

Note: by theorem, all valid proofs must be writable as sum-of-squares inequalities based on combinations of necessary and sufficient interpolation inequalities (hence any use of another inequality, while also leading to an upper bound on rate, may destroy its tightness!)

Worst-case gradient method with constant $\frac{1}{L}$ stepsize

Apply PEP to N steps of gradient method with $\frac{1}{L}$ stepsize:

1. Notice numerical values of worst-case match formula

$$w^*(1, 1, N) = \frac{1}{2N+1}$$

(easy to see for small N , then verified for large N)

2. Guess from (factor of) Gram matrix the values of gradients g_i
Matrix is rank one, interpolating values leads to univariate piecewise linear-quadratic function f_ρ (Huber) with $\rho = \frac{1}{2N+1}$:

$$f_\rho(x) = \begin{cases} \rho|x| - \frac{\rho^2}{2} & \text{when } |x| \geq \rho \\ \frac{x^2}{2} & \text{when } |x| \leq \rho \end{cases}$$

Easy to check performance of method on f_ρ (starts with $x_0 = 1$, then $x_i = 1 - i\rho$), which establishes $w^* \geq \frac{1}{2N+1}$

Worst-case gradient method with constant $\frac{1}{L}$ stepsize

3. To prove upper bound on w^* , it is sufficient to exhibit a dual feasible solution with same value
To describe a dual solution it is actually sufficient to give multipliers $\lambda_{i,j}$ for each interpolation inequality $i, j \in I$

Again guess from numerical values ; e.g. for $N = 1$:

$$\lambda_{0,*} = \lambda_{1,*} = \lambda_{1,0} = \frac{1}{2} \quad \text{and all other multipliers zero}$$

4. Finally one needs to prove (analytically) that the above set of multipliers is feasible (hardest part is proving the slack matrix is positive semidefinite), leading finally to $w^* \leq \frac{1}{2N+1}$

Proof from dual multipliers

Actually SDP formulation/machinery is not needed to derive a **proof of the upper bound**: one can alternatively and equivalently multiply the **interpolation inequalities** by the **dual multipliers** and simplify the resulting expression

For example when $N = 1$, one can check that the following identity holds when $x_1 = x_0 - \frac{1}{L} \nabla f(x_0)$:

$$\begin{aligned} & f(x_1) - f(x_*) \\ = & \frac{L}{6} \|x_0 - x_*\|^2 \\ - & \frac{1}{2} \left[f(x_0) - f(x_1) - \nabla f(x_1)^\top (x_0 - x_1) - \frac{1}{2L} \|\nabla f(x_0) - \nabla f(x_1)\|^2 \right] \\ - & \frac{1}{2} \left[f(x_*) - f(x_0) - \nabla f(x_0)^\top (x_* - x_0) - \frac{1}{2L} \|\nabla f(x_0) - \nabla f(x_*)\|^2 \right] \\ - & \frac{1}{2} \left[f(x_*) - f(x_1) - \nabla f(x_1)^\top (x_* - x_1) - \frac{1}{2L} \|\nabla f(x_1) - \nabla f(x_*)\|^2 \right] \\ - & \frac{L}{6} \left\| x_0 - x_* - \frac{3}{2L} \nabla f(x_0) - \frac{3}{2L} \nabla f(x_1) \right\|^2 - \frac{L}{8} \|\nabla f(x_0) - \nabla f(x_1)\|^2 \end{aligned}$$

(hardest part is grouping of **slack terms into squares**, equivalent to proving semidefiniteness of slack ; proof is not necessarily unique!)

Gradient method, L -smooth convex function, constant step-size $\frac{h}{L}$

Worst-case rate for final iterate accuracy, for any $h \in [0, 2]$

$$\max f(x_N) - f^* = \frac{LR^2}{2} \max \left(\frac{1}{2Nh + 1}, (1 - h)^{2N} \right)$$

We actually know

- ▶ analytical expression for worst-case performance
- ▶ analytical expression of worst-case function (Huber/quadratic)
- ▶ analytical expression of dual multipliers

Even then a rigorous sum-of-squares proof can be hard to obtain!

- ▶ Drori and Teboulle 2014 for $h \leq 1$ (6 pages of tedious algebra)
- ▶ Teboulle and Vaisbourd 2023 for $h \leq 1.5$
- ▶ Rotaru, G., Patrinos 2024, Kim 2024 for $h \leq 2$

Reflections

- ▶ Automated procedure to **compute** worst-case rates
But: is a PEP proof the final goal?
- ▶ More than once, these steps actually happened
 1. Numerical rate computed, analytical expression guessed/identified/confirmed
 2. Using insight provided by worst-case/proof, a closer look at rate/proof provides further intuition and new ideas
 3. Result can now be derived in standard way
(and its “PEP” origin becomes unnoticeable!)
- ▶ Even when such simplifications are not found, shouldn't the goal of mathematical optimization theory first and foremost to increase our understanding?

To conclude: a central open question

- ▶ Can we **design** first-order methods using PEP?

Main difficulty: considering method coefficients to be variable (e.g. stepsizes h_k) destroys convexity of PEP formulation

- ▶ Come this afternoon at **3:15PM** (same place) to learn how to do that!